

# 13 Tribology

## 13.1 General remarks

### 13.1.1 Definition

Tribology is the science of friction and wear. It thus concerns all mechanisms.

In the case of two micromechanical moving parts friction and wear will of course depend on the nature of the bodies, the type of material used and the finish of the surfaces (*fig. 13-1*).

They will also very much depend on:

- ES the **surrounding medium** (lubricant, air, humidity, pollution),
- ES **mechanical conditions** (the load, speed and frequency of contact).

The degree of friction and wear can be strongly influenced by unsuspected parameters.

For example, the interaction time between an oil and a given surface can affect both friction and wear.

### 13.1.2 Surfaces

When two parts move in relation to each other and are in contact their surfaces slide one upon the other. These surfaces constitute discontinuity of the body.

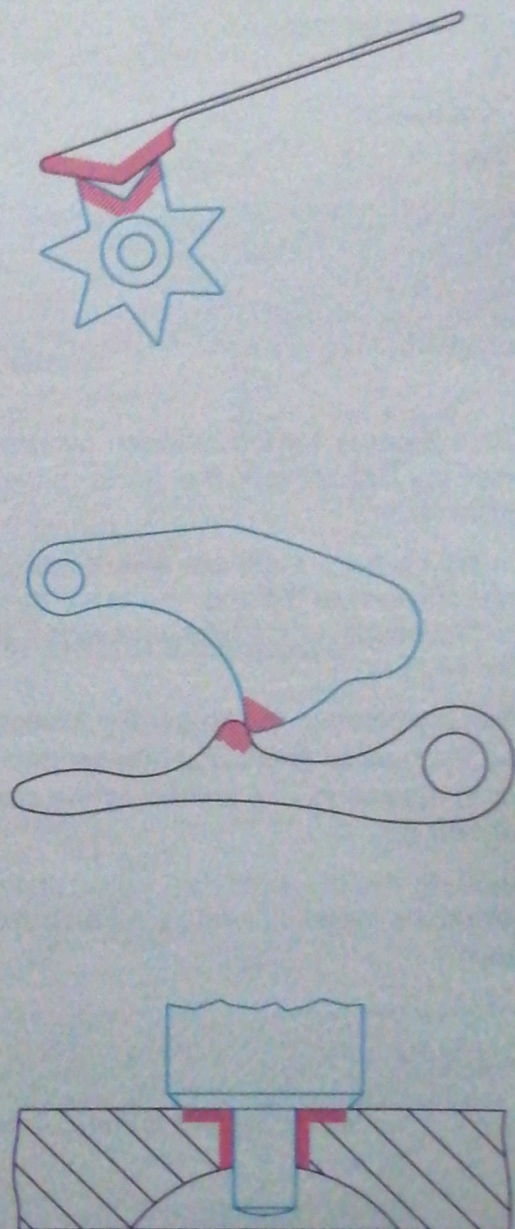


fig. 13-1



Within the material the atoms in effect make up one body. But at the surface they are only partially linked; consequently, they easily react with their surroundings (fig. 13-2).

#### Key

- 1 physicosorbent layer (gaseous)
- 2 chemisorbent layer (gaseous layer which modifies the structure of the metal as a result of a chemical reaction)
- 3 organic molecules
- 4 natural oxide
- 5 cold-worked, contaminated zone
- 6 orientated zone
- 7 intact metal

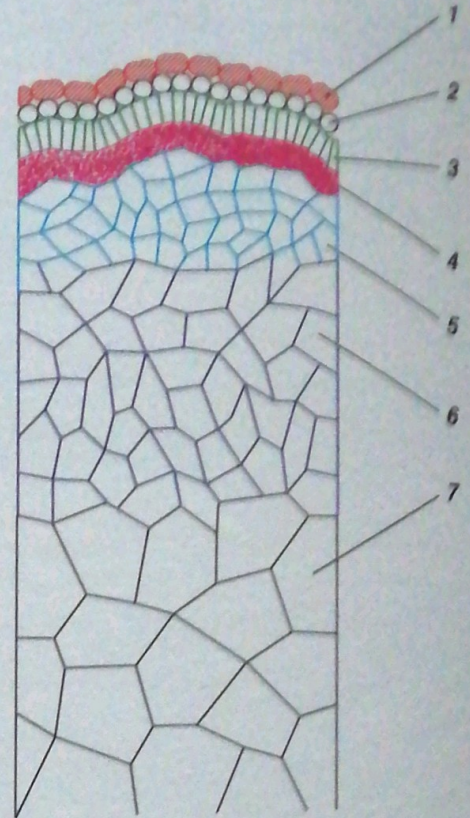


fig. 13-2 : The different layers at the surface of a metal (specular state).

### 13.1.3 Coefficient of friction

When there is friction between two parts there is never contact between the whole of their geometrical surfaces.

On the contrary, there are always many rough sections which may together represent as little as one ten-thousandth of the total geometrical surface area (fig. 13-3).

The ratio between the **tangential friction force ( $F_F$ )** and the **normal force ( $F_N$ )** between two parts moving in relation to one another is the **coefficient of friction ( $\mu$ )**.

Basically, this coefficient is constant and depends on neither the speed, nor the geometrical surface or the load.

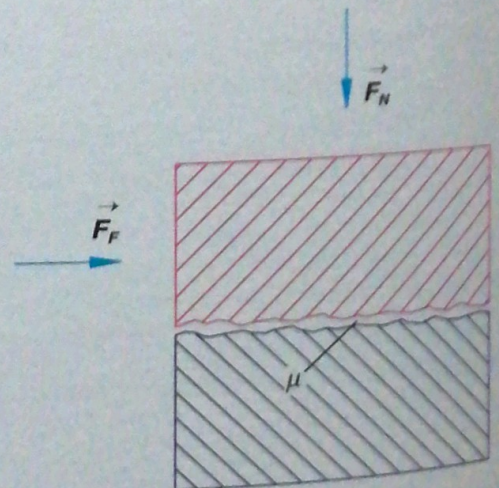


fig. 13-3

#### Key to formula

- $F_N$  normal force pressing one surface to another [N]
- $F_F$  friction force [N]
- $\mu$  coefficient of friction

$$F_F = \mu \cdot F_N$$



When the mechanism is at rest and is set in motion very slowly the coefficient of friction is called a **static coefficient**.

When friction is measured during continuous movement, be it even slow, the coefficient is called a **dynamic coefficient**.

The energy lost through friction is used to shape, break down or juggle the molecules in the lubricant, to break adhesion through roughness and elastic or plastic deformation.

If there is sufficient energy at the point of contact an increase in temperature will be seen; this temperature rise may be considerable if the material is a poor conductor of heat. A high temperature increases the speed of the reaction with agents in the surrounding medium, which leads to:

**oxidation, the formation of salts, carbides and rust, as well as deterioration of the lubricant, if there is any.**

In watchmaking the dimensions of the pivots must be such that the normal load at the point of contact is not sufficient to cause wear in the form of metal fatigue, which will lead to flaking and scratching of the surface. This will happen if the diameter of the pivot is too small in relation to the load it bears.

### 13.1.4 The tribometer

A tribometer is an instrument used for studying the phenomena associated with tribology.

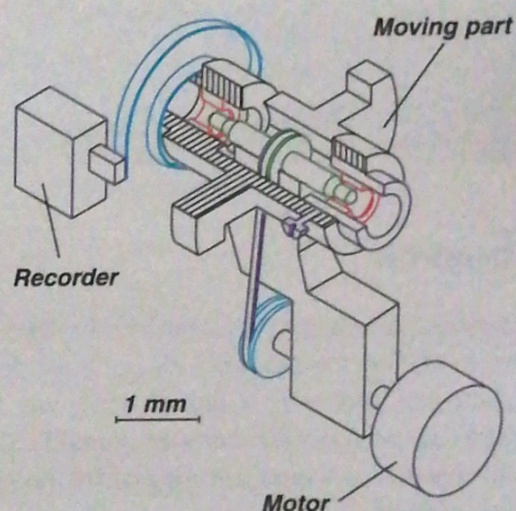
In contrast with other measuring instruments whose only function is to measure, a tribometer at the same time **causes** the phenomenon of friction which is to be measured; this gives rise to a complex variety of methods.

There exist, for example:

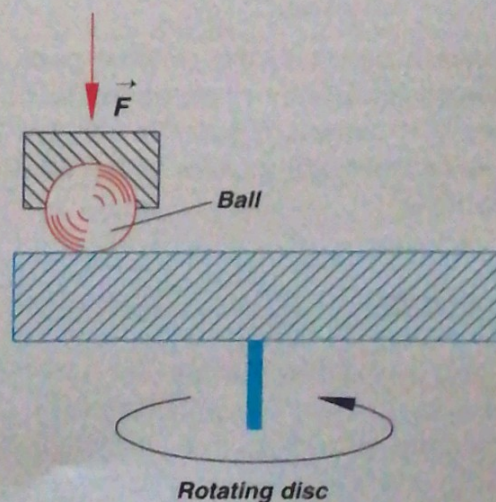
- ▣ precise tests involving ball-disc tribometers for measuring friction and wear (*fig. 13-6*),
- ▣ tests which are quantitatively less precise involving instruments which simulate very accurately the mechanism under scrutiny.



*fig. 13-4 : A rusted pinion.*



*fig. 13-5 : The microtribometer.*



*fig. 13-6*



## 13.2 Wear

Wear occurs when material is removed from one or other of the parts which are in contact, or from both.

The **rate of wear** is the volume of material removed by unit of slide. It is expressed as  $\text{m}^3/\text{Nm}$  or  $\text{m}^2 \cdot \text{N}^{-1}$ .

The international reference unit is  $10^{-15} \text{ m}^2 \cdot \text{N}^{-1}$ . Above this value lubrication is poor, while below it lubrication is adequate. In watchmaking a value of under  $10^{-17} \text{ m}^2 \cdot \text{N}^{-1}$  is aimed at.

### 13.2.1 Different types of wear

#### Abrasion

Abrasion is used in machining: in turning, milling and grinding, for example. A body made of a hard substance and with a sharp edge will wear away softer material where there is contact. Such a body may move freely between two parts, as when rubies are polished.

In the case of mechanical friction wear can occur when the surface of the hardest body is too rough or if particles which are harder than the original body become detached.

Abrasion occurs in the external parts of a watch. Abrasion may occur when the wearer brushes his or her watch against a wall (*fig. 13-7*) or if fine abrasive particles are present in the fibres of clothing (*fig. 13-8*).

#### Erosion

A fluid carries abrasive particles which hit and wear away the surface. This phenomenon is used in **sand blasting**. In matting the impact of the granules makes minute depressions on the surface and lends it certain properties (*fig. 13-9*).

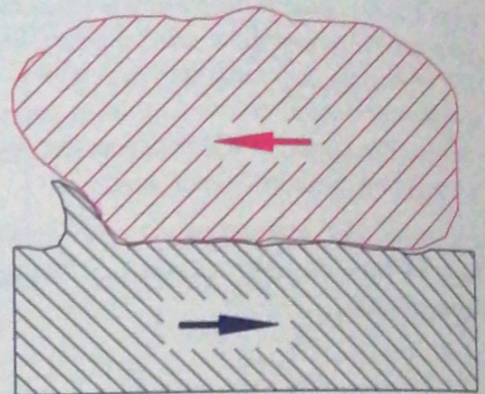


fig. 13-7

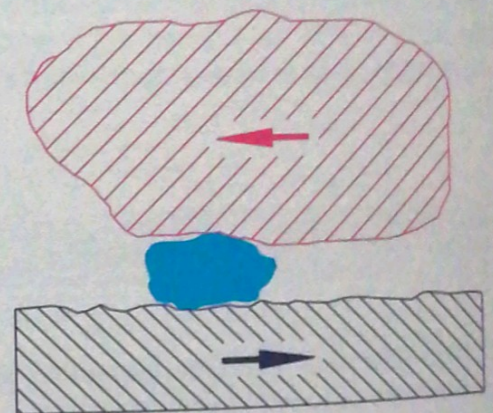


fig. 13-8

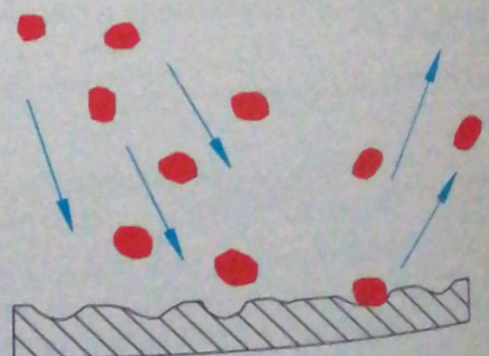


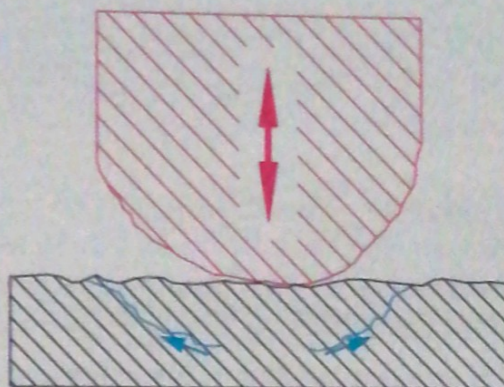
fig. 13-9



## Fatigue

Under extreme or varied periods of stress (*fig. 13-10*) the surface of a material will finally start to flake (*fig. 13-11*).

Cracks form under the surface and eventually reach the outermost layer, whereupon a flake will become detached, making contact even more abrupt. From this point on the damaged section, which bears an increased load, will deteriorate very quickly.

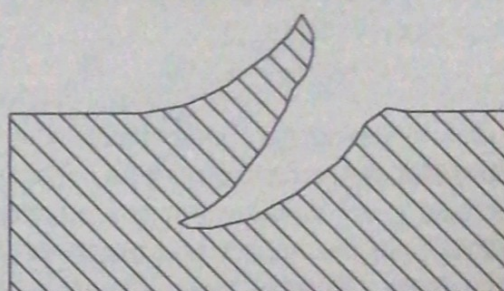


*fig. 13-10*

## Chemical wear

On at least one of the parts the surface will change in reaction to the environment.

This modified layer is worn away by friction and another will form in its place. The material is gradually consumed in this way (*fig. 13-12*).

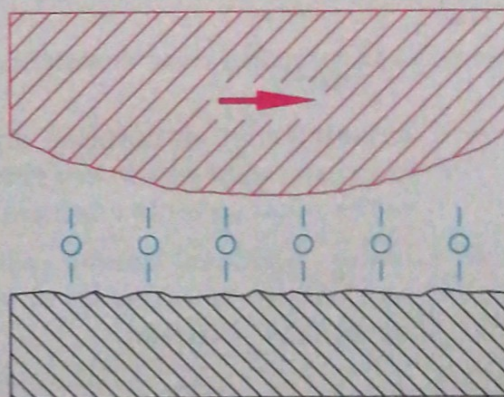


*fig. 13-11 : Flaking.*

## Adhesive wear

This is the most common form of wear in micro-mechanics. When there is not enough lubricant, **microbonding** occurs between the parts which are sliding one over the other (*fig. 13-13*).

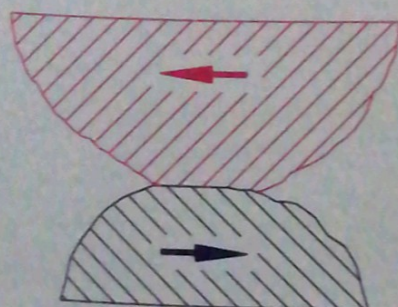
These unions are broken through the mechanical movement and part of one surface may stick to the other, or a piece may become totally detached and thereafter pollute the lubricant (*fig. 13-14*).



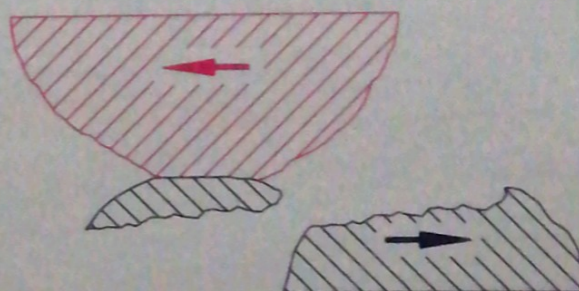
*fig. 13-12*

## Combined wear

In practice several different types of wear will normally occur successively or simultaneously.



*fig. 13-13*



*fig. 13-14*



### 13.2.2 Friction and wear in watchmaking

In machines where there is enough room excellent solutions can be adopted to reduce wear and the coefficient of friction, especially for rotating movements (*fig. 13-15*).

This includes ball bearings and replaceable oil films in plain and porous (sintered) bearing blocks.

Such solutions are generally used very little if at all in watchmaking because of the lack of space, the relative precision which must be achieved and enormous local pressure.

In **watchmaking** the speed of slide is so low that an active film of lubricant cannot be considered, except perhaps in the escapement of a mechanical watch. In this case the impact and breaking of the film at each contact must be taken into account.

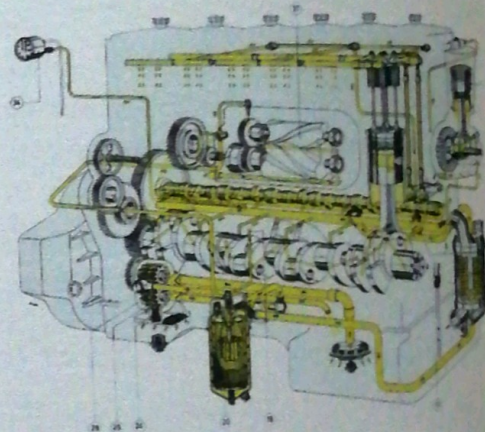
In watches lubricants must remain effective without being changed for many years, only a few micrograms being used at each lubrication point.

**The ways of reducing wear in watches include:**

- ☞ ensuring that all surfaces are perfectly clean and applying a very small quantity of the most suitable lubricant;
- ☞ using surfaces coated with a self-lubricating, hard, soft, composite or successive material.



*fig. 13-15*



*fig. 13-16 : Lubrication diagram for a DKL engine (Saurer).*



*fig. 13-17 : Lubrication of the pallet stone.*



## 13.3 Cleaning

### 13.3.1 Definition

Cleaning consists of removing all undesirable substances from a surface as thoroughly as possible and without damaging it.

Such substances may include:

- **solid particles:** sand, dust, fragments of material, abrasive elements, oxides, etc.,
- **residues:** solidified oil, dried grease, polishing paste, etc.,
- **liquids:** water, various solvents, oil, cleaning products, etc.

### How to clean ?

- solid particles can be removed mechanically by brushing, mechanical vibration, blowing or ultrasonic vibration (fig. 13-19),
- some substances can be dissolved in suitable liquids,
- soap, diluted with water, will act chemically on certain foreign bodies and dissolve them,
- heat can be used to accelerate removal.

### 13.3.2 Cleaning agents

Whatever products are used it is essential that they are compatible and will react with each other to leave absolutely no residue. Water cannot be eliminated with benzene, for example, but alcohol will do the job.

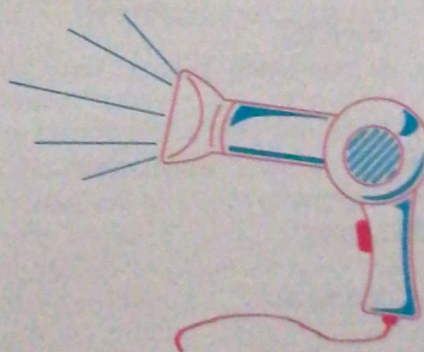
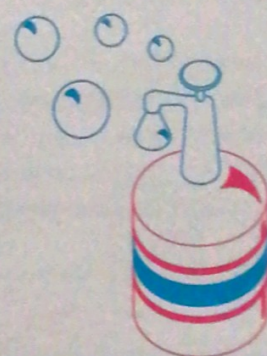
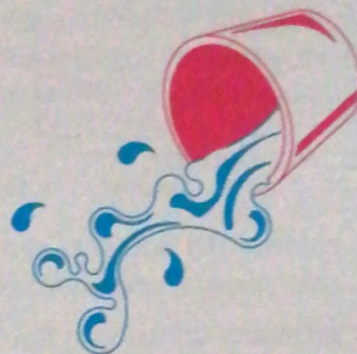
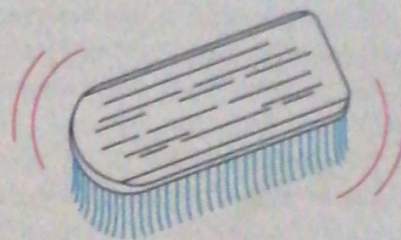


fig. 13-18 : How to clean ?

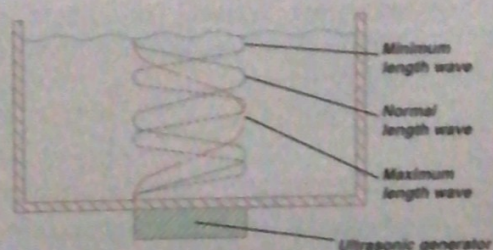


fig. 13-19 : Frequency-scanning ultrasonic bath.



Whatever cleaning agent is used, a solvent or a detergent, we must first ensure that it is a suitable product for removing the dirt in question.

Various ranges of very effective and popular products are complicated to use because certain precautions must be taken to avoid polluting the atmosphere, or because they are either environmentally damaging or toxic for humans. The installations which are necessary for using such products are expensive and can only be considered if they are to be used for a large number of pieces.

Such products are not inflammable and have a low boiling point, which means they are suitable for rinsing; unfortunately they evaporate rather quickly.

These products include:

- ☞ **chlorinated solvents** (toxic) such as perchlorethylene and trichlorethylene,
- ☞ **products containing fluorine** (pollutants), such as freon.

There are several other popular products which, however, have the disadvantage of being inflammable, especially when heated, since the vapour may explode. As these products evaporate very quickly it is easy to dry the pieces.

These include:

- ☞ **petroleum derivatives** (benzine, etc.).
- ☞ **alcohol.**

In view of all the points mentioned above we have to resort to products which clean efficiently but are difficult to dry, especially at the bottom of blind and threaded holes.

Such products are:

- ☞ **water-based solutions**, i.e. soaps or detergents diluted in water and rinsed off with **hot water**. This helps evaporation during drying, which can only be done with hot air.

**Disposal of all such products is covered by special legislation.**

## POISON

### Classes 1 and 2

Beware ! Not to be taken internally.  
Keep under lock and key. Caustic. Avoid contact with skin and eyes. Do not inhale vapour. Use only in the open air or by an open window. Wear protective glasses.



### Class 3

Beware ! Toxic substance. Not to be taken internally.  
Keep out of reach of children and non-competent persons. Avoid contact with skin and eyes.

### Classes 4 and 5

Not to be taken internally.  
Keep out of reach of children and non-competent persons. Inflammable. Do not inhale vapour. Use only in the open air or by an open window.

fig. 13-20 : Swiss toxicity classes.

**Ammonia**

2

**Trichlorethylene**

4

**Benzine**

4

**Alcohol**

5

fig. 13-21 : Toxicity class of different cleaning agents (Swiss system).

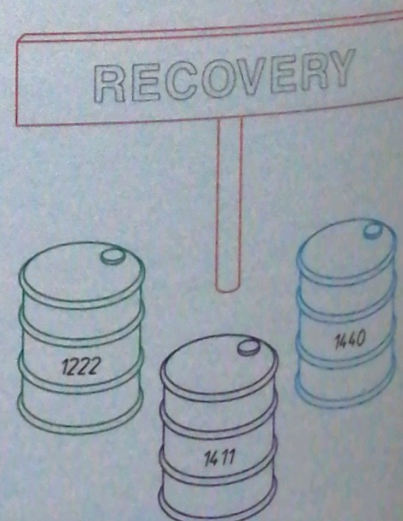


fig. 13-22 : Recovery of cleaning agents.



In order to eliminate the deposit of residues when a piece is dried it is necessary to rinse it with de-ionised water. This has the disadvantage of causing corrosion in certain metals which are not protected by electroplating, as in burnished pivots, for example. Certain precautions must therefore be taken.

For rinsing we can use absorptive agents which allow water to be evacuated more easily. There is a disadvantage here, however, in that they leave an invisible and often undesirable film on the surface of the part in question.

### 13.3.3 Cleaning machines

The design of such machines will depend on the cleaning agents used.

#### First type

Containers are filled with different solutions which can be heated to the desired temperature. The parts to be cleaned are either placed in a basket or suspended in a frame and are passed from one container to the next. The solutions, or the parts in the baths, are agitated by rotation, mechanical or ultrasonic vibrations. Hot air is used to dry the parts.

#### Key

- 1 cleaning solution
- 2 rinsing solution
- 3 heating element
- 4 ultrasonic generator
- 5 ventilator
- 6 basket or frame (rotating or not)

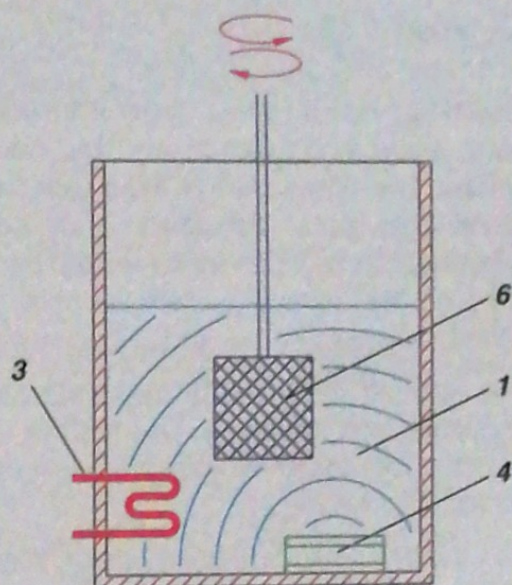


fig. 13-23 : Cleaning.

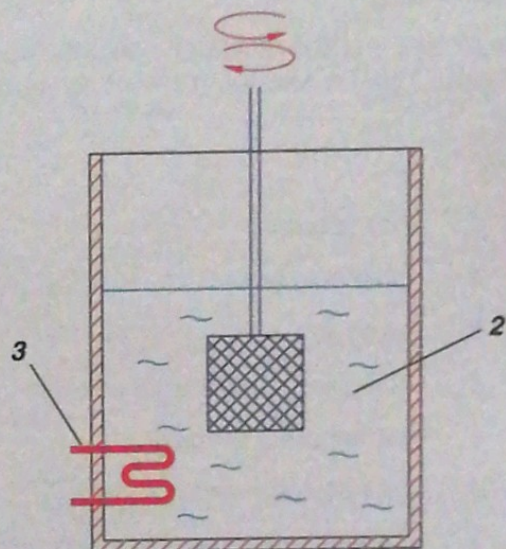


fig. 13-24 : Rinsing (there may be several rinsing baths).

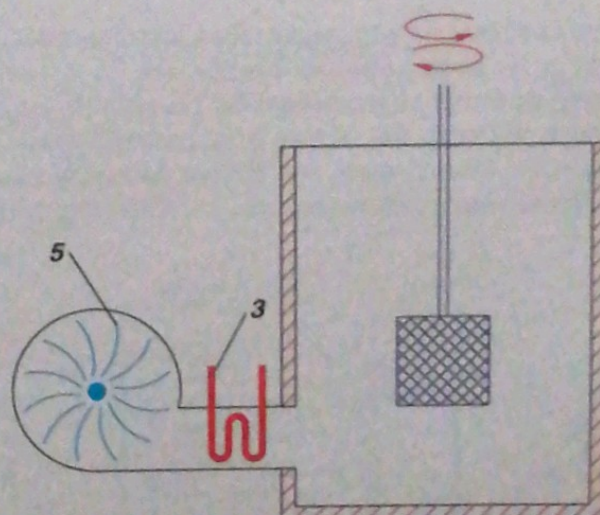


fig. 13-25 : Hot-air drying.



## Second type

These are made to be used with solvents which have a low boiling point and therefore evaporate quickly. The upper part is fitted with tubes through which cold water circulates. They condense the vapour which escapes from the bath and the distilled form of the solution, which is pure, is thus obtained.

The solution is heated to clean the parts and any impurities are dissolved. The bath is ultrasonically agitated (fig. 13-26). After a first rinsing the parts are cooled (fig. 13-27).

The second rinsing is with vapour from the boiling solution which condenses and flows down over the parts. They are then raised slowly until they are out of the vapour zone, by which time they are dry (fig. 13-29).

To ensure perfect results these machines work with a closed circuit and must be totally sterile.

### Key

- 1 hot solution
- 2 cold solution
- 3 vapour
- 4 (distilled) condensed solution
- 5 ultrasonic generator
- 6 heating element
- 7 cooling coil containing cold water

## Third type

Electrochemical cleaning involves alkaline solutions. The parts suspended in frames are in either an anode or a cathode position. By passing an electric current through the water it is broken down into hydrogen and oxygen, the former reducing oxides while the latter oxidises organic particles (fig. 13-28).

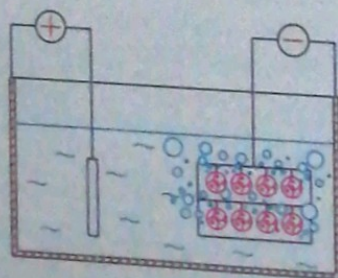


fig. 13-28

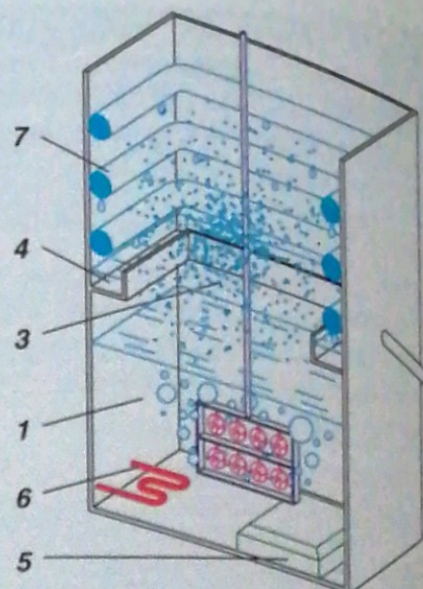


fig. 13-26 : Cleaning.

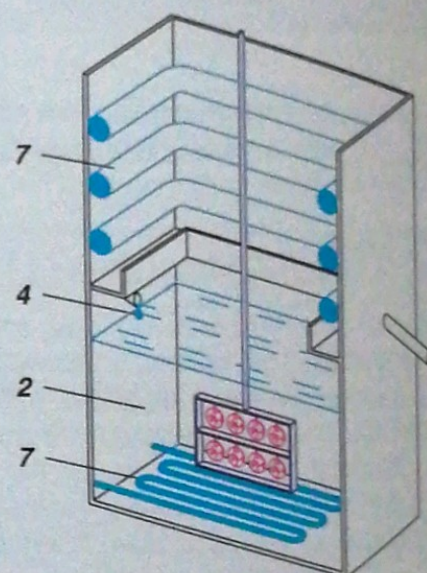


fig. 13-27 : First rinse.

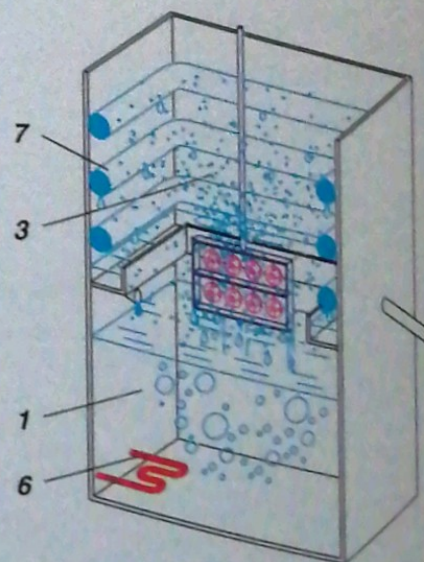


fig. 13-29 : Second rinse and drying.



## 13.4 Lubrication

### 13.4.1 Definition

Lubrication is the action of inserting a product between two parts which are in contact with each other: the **lubricant** reduces the coefficient of friction and wear (fig. 13-30). There are three forms of lubricants: solid, liquid or gaseous.

#### The Stribeck curve

The curve representing friction as a function of speed, pressure and viscosity in a lubricated bearing was first drawn by **R. Stribeck** in 1903 and is thus called the Stribeck curve.

Each lubricant produces a particular curve under precisely defined conditions.

#### Key

|            |                         |
|------------|-------------------------|
| $\mu$      | coefficient of friction |
| $p$        | pressure                |
| $v$        | linear speed            |
| $\omega$   | angular speed           |
| $\eta$     | viscosity               |
| $\epsilon$ | clearance               |

The curve has three zones:

- 1 the **oily or limit of friction zone**: the effects of adhesion at rough points are seen less and less, but the speed is too low for the wedge of oil to have any effect. The mean distance between rough patches increases gradually to  $0.1 \mu\text{m}$ .
- 2 The **elasto-hydrodynamic zone**: this is the zone where the coefficient of friction is minimal. There is still some contact between rough patches if the load is above a certain value. The speed is still too low for a wedge of oil to form.

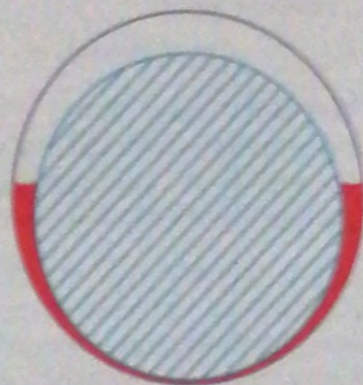


fig. 13-30

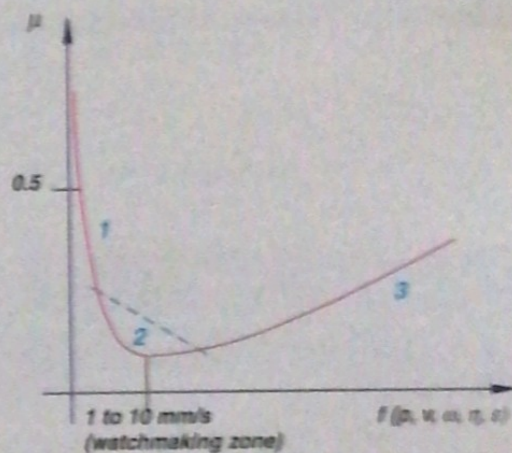


fig. 13-31 : The Stribeck curve.



- 3 **The hydrodynamic zone:** the speed is sufficient for a wedge of oil to prevent direct contact between rough patches. The coefficient of friction is minimal at first. As the speed increases the load borne exceeds the value required and more and more energy is lost as the layers of oil are sheared. In this zone there is minimum wear since there is no contact between rough patches except when the mechanism starts or stops.

The shape of the curve serves only as an approximate and rudimentary interpretation of the phenomenon. In relation to watchmaking, only zones 1 and 2 are important.

The diagram opposite (fig. 13-32) gives an idea of the speeds involved in bearings for watches where the balance oscillates at a frequency of between 3 and 4 Hz (21,600 vib/h and 28,800 vib/h respectively).

The limit for hydrodynamic lubrication is reached with a balance which oscillates at 4 Hz. It is true to say that watches are lubricated entirely in the oily zone with perhaps some elasto-hydrodynamic effects on the pivots of the fastest moving parts.

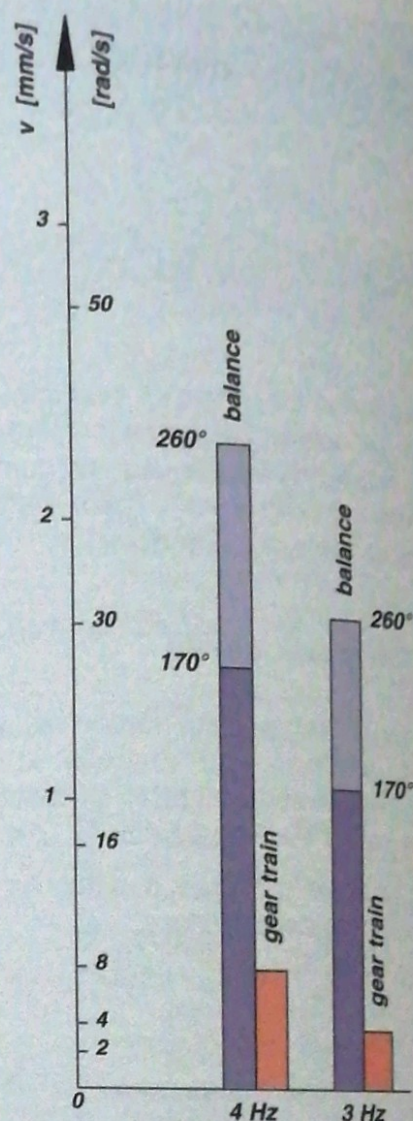


fig. 13-32

### 13.4.2 Lubricants used in watchmaking

**What qualities should a lubricant have ?**

- 1 chemical stability
- 2 non-corrosive properties for components
- 3 non-creep stability
- 4 low temperature stability
- 5 low volatility
- 6 stable and low coefficient of friction
- 7 prevention of wear
- 8 maintaining a stable and high amplitude irrespective of the humidity

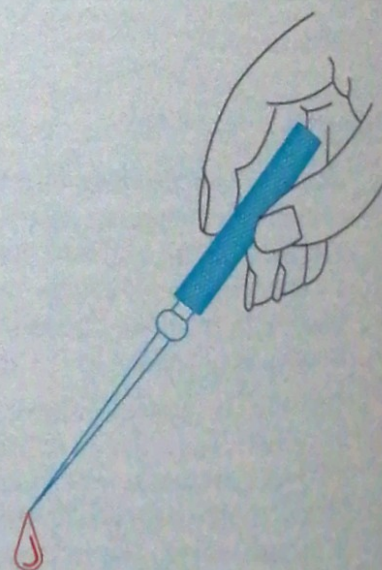


fig. 13-33 : Many qualities are required of one tiny drop of oil.



# What lubricants are available to the watchmaker ?

## Classical oils

These are either vegetable (olive oil) or animal (neat's foot oil) in origin. Their lubricative properties are excellent and they would seem to be the ideal lubricants for watchmaking.

Unfortunately, when they come into contact with air and through the catalytic effect of metals containing copper, they tend to oxidise easily and therefore lose their effectiveness rather quickly. Attempts have been made to improve them by more careful refining and by mixing them with mineral oils. Despite all the research which has been done so far it has not been possible to produce a degree of stability anywhere near that offered by synthetic oils.

## Synthetic oils

Synthetic oils were first produced in 1950 by a complex combination of chemicals. They can be categorised into several groups. The first synthetic oils produced satisfied the first 5 requirements listed above. In contrast, no-one seemed interested in the lubricative properties per se which are listed under 6, 7 and 8. Studies in this connection were only carried out at the end of the 1970s.

These synthetic oils have been constantly improved with the aim of finding a product which lasts for as long as possible.

## Grease

This is a combination of oil and a fibrous soap which serves as a sponge to retain the oil. There is also a type of grease used in watchmaking which is made up of oil and molybdenum disulphide ( $\text{MoS}_2$ ) powder as a binding agent.

Grease has to be used very carefully. It must be perfectly homogenous before it can be used and should thereafter separate easily: the binding agent ensures that the product remains in place and the oil acts as a lubricant. As far as the escapement is concerned, a grease must not produce micro-droplets when exposed to shocks or when mechanical contact is initiated.

## Self-lubricating materials and coatings

Numerous studies have been carried out and are still being done without any miracle solution being discovered so far.

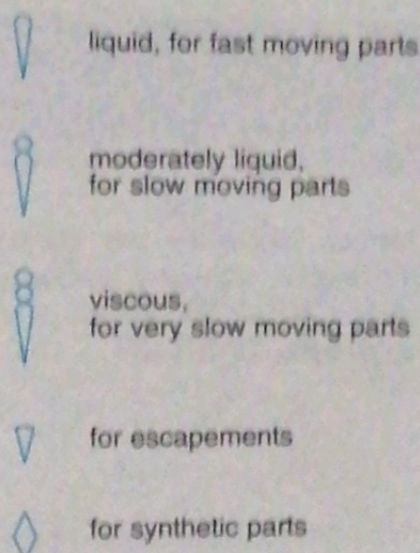


fig. 13-34 : Different types of oil.

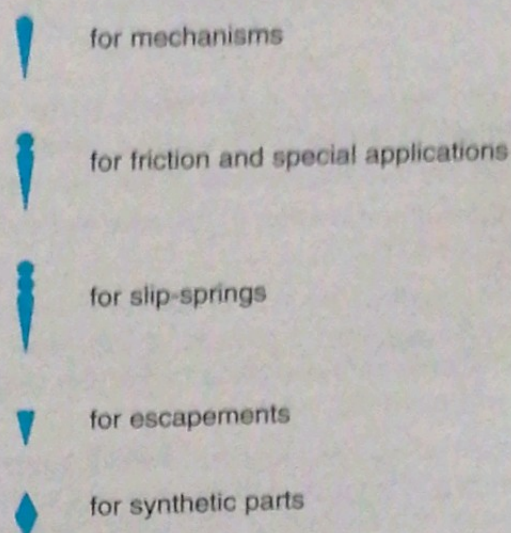


fig. 13-35 : Different types of grease.

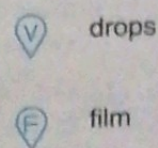


fig. 13-36 : Forms of lubrication.

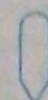


fig. 13-37 : Pre-lubricated state.



fig. 13-38 : Do not lubricate.



### 13.4.3 Maintaining oil on a surface

A liquid whose properties are identical in every direction (isotropic), without a tendency to polarise will form drops whose equilibrium depends on the surface tension (fig. 13-39).

#### Key

- $\sigma_s$  apparent surface tension of the support
- $\sigma_h$  surface tension of the oil
- $\sigma_i$  interfacial tension between the oil and the support (the sum of  $\sigma_s$  and  $\sigma_h$ )

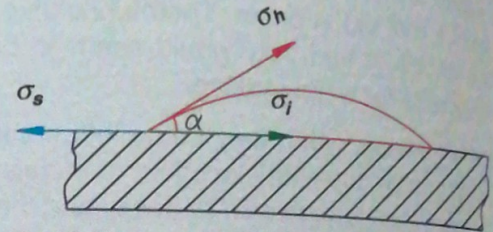


fig. 13-39

It can be stated in simplified terms that **oil will remain in place if its surface tension is greater than the surface tension of the support** (fig. 13-40).

If the surface tension of the oil is too low it will creep (fig. 13-41).

If the surface tension of the oil is far too great the droplet will form a globule. It then does not *adhere* sufficiently to the surface and at the slightest jarring it will move (fig. 13-42).



fig. 13-40

When parts are cleaned, a watchmaker will sometimes add a product to the final rinse which leaves a film on the surface of the parts. This product, called an **epilame**, will change the apparent surface tension of the support in one direction or the other. In fact it either *binds* the oil so that it does not creep or it *repels* the oil to ensure that it stays in certain defined areas.



fig. 13-41

The same theory applies in reverse to cleaning agents. In this case a watchmaker deliberately chooses softeners which have a low surface tension and will easily infiltrate into the tiniest recesses to remove impurities.

Soap is an example of such a *water-softening* product. It is able to change the properties of water by enabling it to moisten a surface, our hands or a cloth more thoroughly. In this way it is easier to remove impurities.

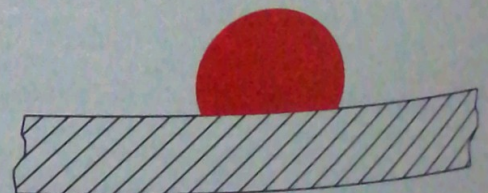


fig. 13-42



### 13.4.4 Choosing a lubricant

In view of the complexity of the problems to be solved a watchmaker is obliged to pay close attention to:

- the recommendations of suppliers of lubricants for watchmaking.
- technical information supplied by watch-making companies which indicates which cleaning agents should be used, which lubricants are recommended, the quantities to be applied and the lubricating points (fig. 13-43).

**It is essential that oil be applied only to perfectly clean surfaces.**

**It serves no purpose to apply oil to a dirty part or to a part which has already been oiled, as the oil will not stay in place.**

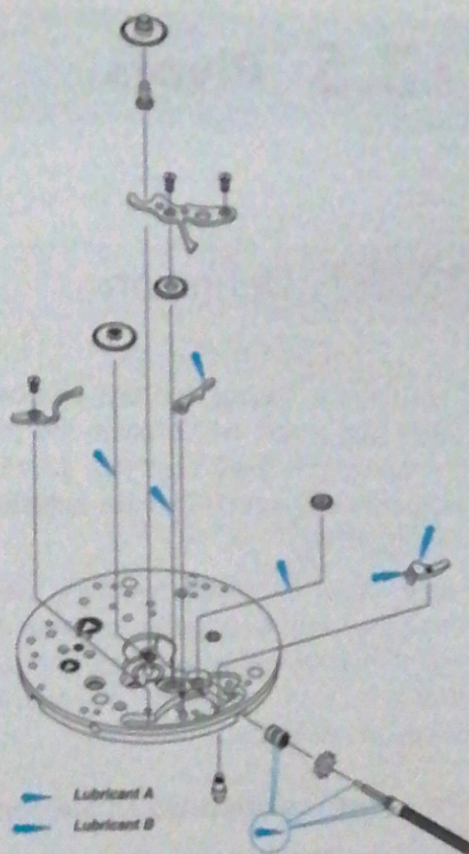


fig. 13-43

### How to test a surface

#### Before applying a lubricant:

By using a particular process involving a range of 6 special oils with a surface tension ranging from 19.3 to 40.8 mN·m<sup>-1</sup>, it is possible to determine the apparent surface tension which should produce the best results.

#### After applying a lubricant:

Prepare small, smooth brass plates which have been electroplated like the watch plate.

Each time the bath or lubricant is changed clean one prepared test plate at the same time as the movement is cleaned. Write the date on the plate and apply a drop of each of the lubricants used (fig. 13-44). Store the plate away from the light and plastic and in a dust-free environment.

Check the state of the lubricants regularly. As long as there is no creep all is well. If one or several of the lubricants show signs of creep or have disappeared (fig. 13-45) the same will have happened in the movement and it will no longer function properly.

The process should therefore be modified.

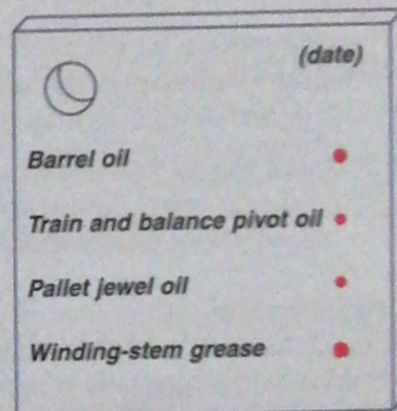


fig. 13-44

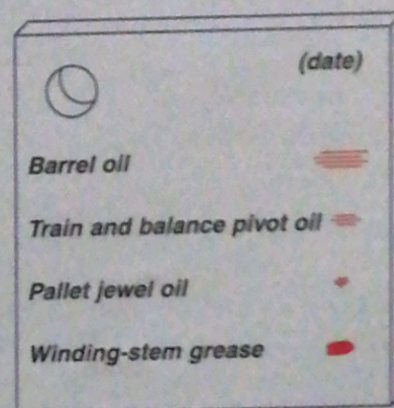


fig. 13-45 : Condition after a certain time.



## 13.5 Pivots

### 13.5.1 Definition

In horology a special solution has been found for wheels and levers which pivot: the pivoting part is either set into a fixed hole (*fig. 13-46*) or a hole is made specially which can be adjusted on a fixed axis (*fig. 13-47*).

In both cases, there are two parts which will come into contact with each other. In order to ensure that the frictional torque is as weak and as constant as possible (to keep losses to a minimum) the following choices must be made:

- ☛ **two materials which are compatible:**
  - polished, hardened or burnished steel with ruby;
  - polished, hardened or burnished steel with brass or bronze;
- ☛ **dimensions** for the pivot and the bearing which are **adapted** to the levels of pressure and with some **play** or clearance;
- ☛ a quantity of **lubricants** which allows for a reserve: this is the function of the small recess called the **oil-sink** (*fig. 13-48*).

Bearings are designed in such a way that a drop of oil or grease will be held in place by capillary attraction as long as possible in order to ensure that the parts are lubricated. Bearings are made out of:

- ☛ **brass:** holes in the plate or bushes pressed into the plate or the bridges;
- ☛ **beryllium bronze:** bushes pressed in;
- ☛ **corundum** (synthetic ruby, ceramic): jewels pressed in to the plate or the bridges.

### A pivot with a bush or a jewel

Bushes are shaped like normal pierced jewels with a cylindrical hole (*fig. 13-49*). Olived (waisted) jewel-holes are made for high quality jewels. This reduces the risk of scratching the pivot owing to the difficulties of aligning it.

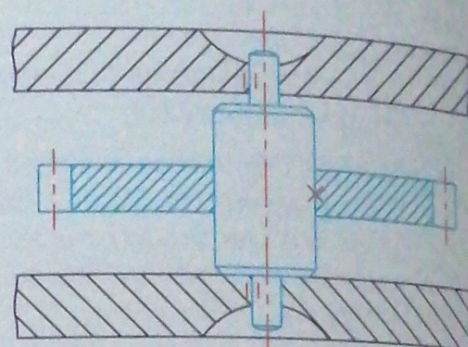


fig. 13-46

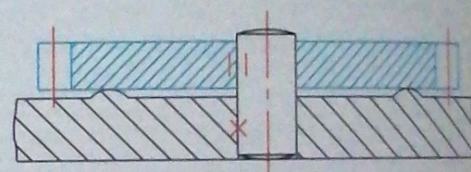


fig. 13-47

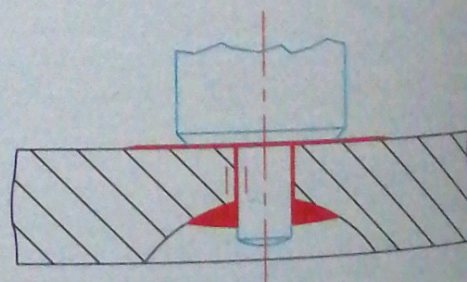


fig. 13-48

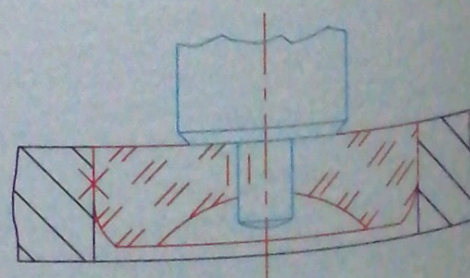


fig. 13-49



## 13.5.2 Other types of pivoting

### A pivot with a jewel and an endstone

In order to reduce friction to a minimum the pivoting of the balance staff and sometimes the escape wheel involves two jewels, the corresponding jewels being fitted in a metal ring called a **setting**. In this case the pivot is **conical** and has no shoulders (fig. 13-50).

The pierced jewel is domed and limits radial movement, while the undrilled endstone limits axial movement. In this case there is no friction on the shoulder but only on the tip of the pivot, which has a very special shape (see chapter on adjusting).

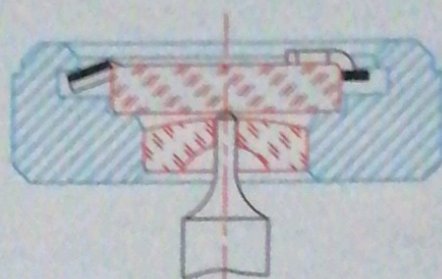


fig. 13-50

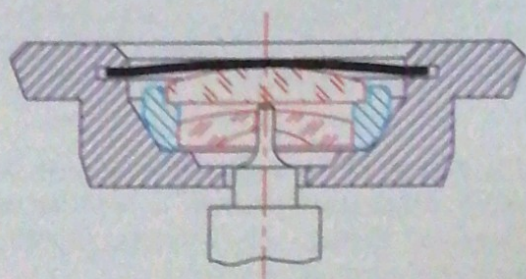


fig. 13-51

### Shock-resistant pivots and settings

Shock-resistant systems have been devised in order to avoid the balance-staff pivots (0.055 to 0.10 mm in diameter) breaking if the watch is jarred violently (fig. 13-51).

The principle here consists of causing the shock to be absorbed by a more robust part of the balance staff than the pivots: it is the pivot shanks which will hit against the body of the device.

In order to achieve this the jewels are fixed in place with a certain degree of elasticity. If the watch suffers no jarring the parts pivot normally. If, however, the watch is jarred the jewels move and allow the pivot shank to absorb the shock (fig. 13-52).

All the jewels subsequently revert to their normal positions.

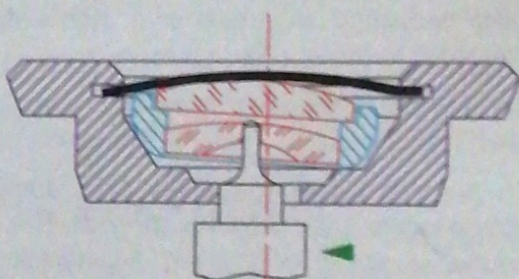


fig. 13-52

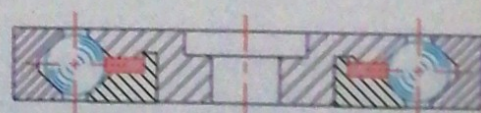


fig. 13-53

### A ball-bearing pivot

Attempts have been made to replace train jewels by bearings with three balls (fig. 13-53). The balls are so small that they do not roll, however, and the result is therefore not satisfactory.

Ball bearings are used in particular for the pivoting of the oscillating weight and certain wheels and pinions in automatic mechanisms.

The use of ball bearings makes it easier to fix the oscillating weight and to increase its stability.

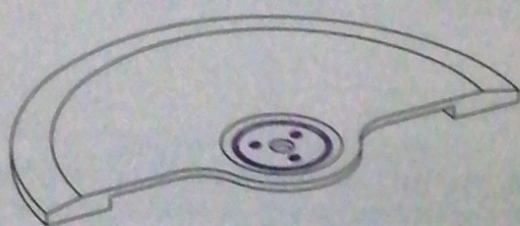


fig. 13-54 : Oscillating weight.



## 13.6 Jewels in watches

### 13.6.1 Historical background

It was the Swiss astronomer and optician **Nicolas Fatio de Duillier**, who was based in London, who first discovered a way of drilling natural rubies so that they could be used as bearings in watches. He patented his invention in **1704** and the French watchmaker De Beaufré subsequently produced the first watch which incorporated rubies.

The history of the *synthetic ruby* goes back to **1892** when **Auguste Verneuil**, a lecturer at the Paris Technical College, discovered a process for manufacturing synthetic stones.

In 1902 Verneuil published a paper on the reproduction of rubies by fusion using an oxyhydrogen blow-pipe. He observed that *from the chemical, physical and crystallographic points of view the properties and molecular structure of a synthetic ruby were identical to those of a natural ruby.*

Only a few years later Verneuil's invention would be developed into a process which is still used today.

In practice synthetic rubies offer some important advantages over natural stones. The man-made stone is purer and more homogenous than its natural counterpart. The imitation in fact surpasses the original, in particular with regard to industrial uses.

### 13.6.2 The manufacture of synthetic corundum

Corundum is made in 3 phases.

#### Phase 1

Ammonium alum is purified by recrystallisation of the double sulphate of aluminium and ammonium (fig. 13-56).

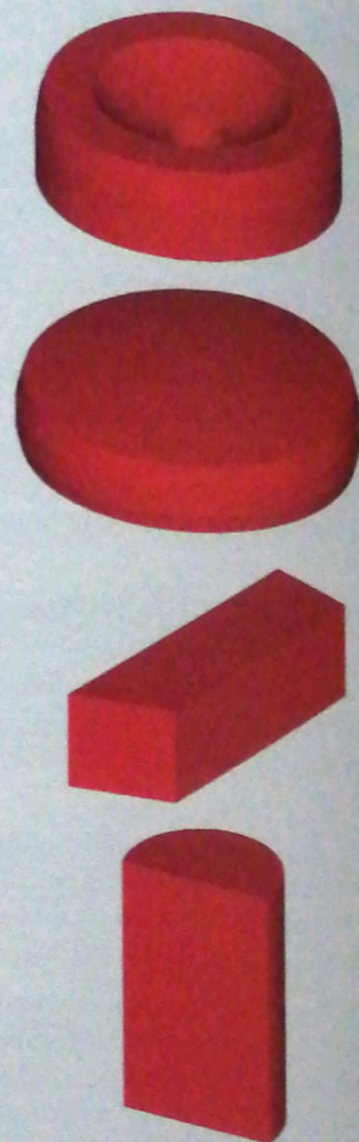


fig. 13-55 : Some different shapes of jewels in a watch.



fig. 13-56



## Phase 2

The purified ammonium alum is calcinated at  $1200^{\circ}\text{C}$ . The resulting product is pure alumina ( $\text{Al}_2\text{O}_3$ ) in the form of small, extremely pure crystals. These crystals are then sieved to give fine particles of  $0.1\text{ }\mu\text{m}$  (fig. 13-57).

## Phase 3

The alumina is then crystallised using a Verneuil blowpipe (figs. 13-58 and 13-59).

A container fitted with a sieve at its base is filled with the alumina powder (3) and inserted into an air-tight cylinder.

A small hammer (1) driven by a cam (2) knocks the container and causes a controlled amount of alumina (4) to fall through the sieve. This substance is sucked into the flame by the oxygen supply (9). At the hottest point (5), where the temperature reaches  $2050^{\circ}\text{C}$ , it melts and crystallises on to a seed crystal placed on a support (7). This forms what is called the *boule* (6).

The boule grows as successive layers of crystals form and eventually achieves a cylindrical shape about 20 mm in diameter and 70 mm long for normal purposes.

As the boule forms, which takes between 4 and 12 hours, the support is lowered so that the upper part of the growing crystal remains in the same zone of the flame (8).

The temperature of  $2050^{\circ}\text{C}$  is achieved and maintained through a supply of oxygen (9) and hydrogen (10).



fig. 13-58



fig. 13-57

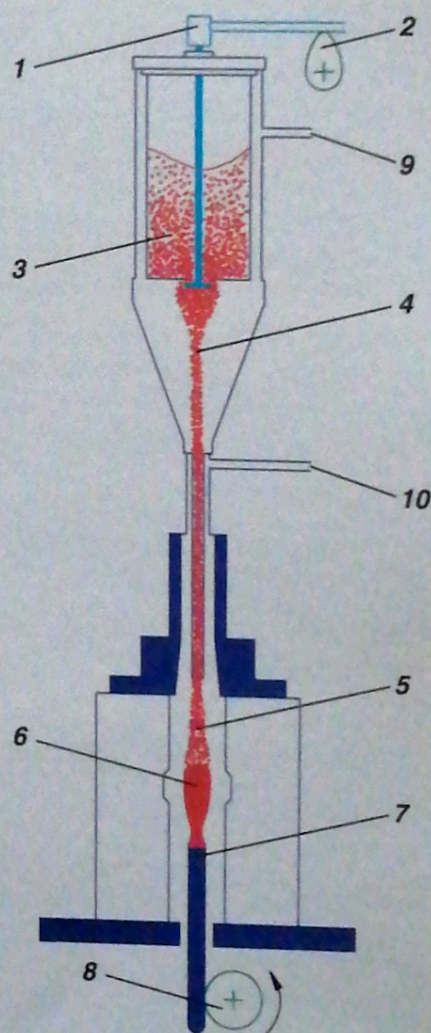


fig. 13-59



### 13.6.3 Synthetic rubies

Rubies, and in particular synthetic rubies, are the hardest stones after diamonds. They can therefore only be worked with diamond tools or powders, or a laser beam.

There is very little wear. A pivot which turns in a ruby bearing will encounter a minimum of friction. The use of rubies in watches therefore provides a considerable increase in precision and life. Another important advantage of the ruby is its homogeneity. It can be highly polished, which again reduces the coefficient of friction.

Synthetic rubies expand hardly at all. They are not affected by changes in temperature, are not corroded by acids and do not conduct electricity.

Like sapphires, rubies belong to the corundum family, which is characterised by its rhombohedral hexagonal crystallisation system. Synthetic rubies are made from extremely pure aluminium oxide (alumina).

The base material of synthetic rubies is:

 **aluminium oxide ( $\text{Al}_2\text{O}_3$ ) or corundum.**

If nothing is added the resulting product is colourless and is in fact white sapphire. If traces of chromium oxide are added the stone will be a beautiful rich red.

The addition of traces of other substances produces further colours: titanium and iron oxide will give blue sapphire, cobalt oxide will produce green corundum and nickel oxide will give yellow sapphire.

#### Characteristics of a synthetic ruby

|                             |             |
|-----------------------------|-------------|
| Hardness                    |             |
| on the Mohs scale           | 9           |
| Hardness                    |             |
| on the Knoop scale (mean)   | 2000        |
| Dry coefficient of friction |             |
| on steel (minimum)          | 0.1         |
| Melting point               | 2050°C      |
| Density                     | 3.99        |
| Refractive index            | 1.76 - 1.77 |



fig. 13-60 : Drilling a watch jewel using a laser beam.



fig. 13-61 : Corundum of various colours.



fig. 13-62 : Different stages in the manufacture of synthetic jewels (boules).



## 13.6.4 Manufacturing jewels

### Preparation

The basic material comes out of the blowpipe in the form of a cylindrical crystal (*fig. 13-63*).

In order to eliminate the internal stresses which arise as the crystal forms and which make it as fragile as hardened (security) glass, the crystal has to be **cleft** along a precise plane, the optical axis, which is determined using polarised light (*fig. 13-64*).

Each part is then sawn into small **plates** or **half-moons** of a fixed thickness, using copper discs whose cutting edge is encrusted with diamond powder (*figs. 13-65 and 13-66*).

The half-moons are subsequently cut into **squares** (*fig. 13-67*) and the corners are rounded off on special machines. This produces **small discs**, or **rough jewels** (*fig. 13-68*).



fig. 13-63



fig. 13-64



fig. 13-65

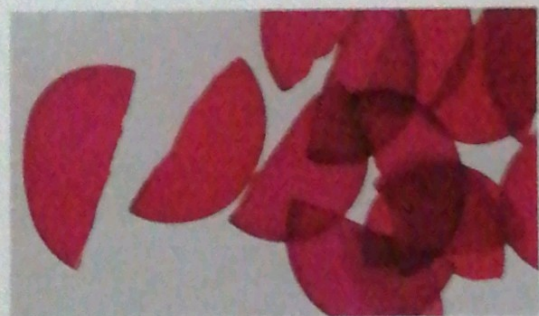


fig. 13-66



fig. 13-67



fig. 13-68



## Finishing

The final form of the synthetic jewel for watchmaking depends on the function it will fulfil within the movement. The range of products thus includes pierced jewels, endstones, pallets and impulse pins.

Since the early 1970s jewels have been **pierced** using a laser beam. The energy transmitted by the beam, which is focused on to the stone, sublimates the matter, ejects fused particles and creates a hole (fig. 13-69).

Part of the matter which has not been ejected partially recrystallises on the surface of the hole. Extremely quick cooling gives the pieces a rough, vitrified appearance with tiny cracks. This zone of fragile, inhomogeneous material has to be removed.

A finished hole with a diameter of 0.09 mm can be produced from a 0.05 mm hole made by a laser beam.

The drilled stones are threaded on to a metal wire covered with diamond paste; this process is called **threading** (fig. 13-70).

The hole is worked until the required diameter is obtained; this step is called **enlarging** (fig. 13-71).

The circumference of the disc is also worked in relation to the hole; this is called **turning** (fig. 13-72).

The oil-sinks are then formed using special machines; this process is called **recessing** (fig. 13-73).

For **grinding** and **polishing** (including chamfering) the stones are cemented on to cast-iron discs with shellac (around 2,500 stones per disc).

Certain jewels, namely those which are to be used for the balance, will need to have olive (waisted) jewel-holes and will thus undergo an **oliving** process.

All these processes involve abrasion which is achieved using diamond powder.

The last steps in the manufacture of watch jewels are **cleaning** and **inspecting** (fig. 13-74).



fig. 13-69



fig. 13-70

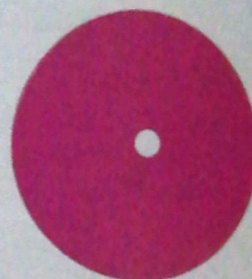


fig. 13-71



fig. 13-72



fig. 13-73



fig. 13-74



## 13.6.5 Other products

Corundum is used not only for watch jewels. This extremely hard and chemically inert substance is in great demand wherever these two qualities are required.

### Watch glasses

Sapphire glasses (fig. 13-75) are scratch-proof and much tougher than normal glass. A sapphire glass is therefore normally used to protect the dial.

Methods have been devised for machining the rim of shaped watch glasses.

Fitting the glass to the bezel requires great precision in the machining of the glass and a plastic gasket or glue is also needed.

### Tips for dial gauges

The sensing tips of dial gauges are often fitted with a sapphire, ruby or ceramic tip which has the advantage of not being subject to oxidation (fig. 13-76).

### Instruments used in medicine and the chemical and space industries

Corundum is also used for special pumps, rubies for lasers, windows with an ultrafine surface to protect the optics of an endoscope (fig. 13-77), prisms, wire guides, or guides for printing styluses, for example.

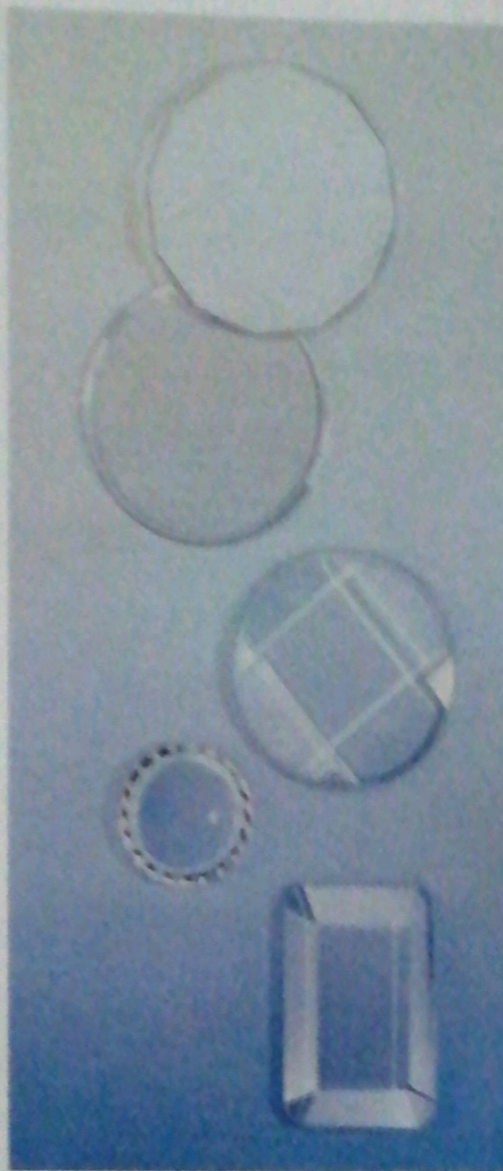


fig. 13-75

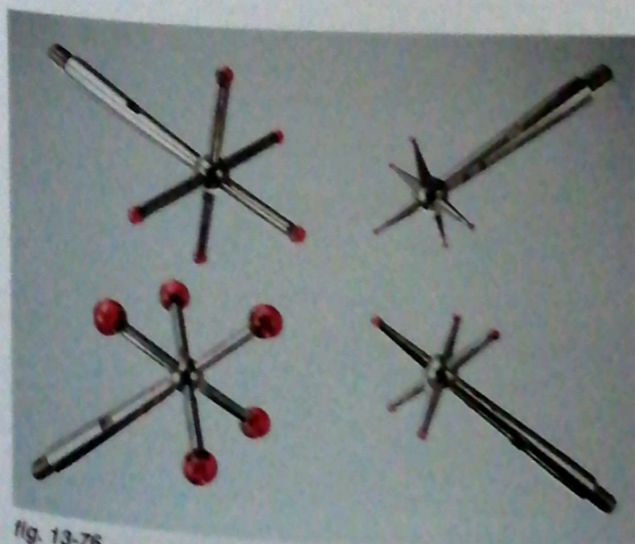


fig. 13-76



fig. 13-77





fig. 13-78 : Grande sonnerie Neuchâtel clock with date; cabinet painted with flowers of various colours. Made by Aimé Billon, La Chaux-de-Fonds, c. 1830. International Watchmaking Museum, La Chaux-de-Fonds (Switzerland).